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Función derivable

Definición 1 (Derivabilidad en un punto). Sea $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ una función, f es derivable en $x_0 \in \mathbb{R}$

$$\iff \exists \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} =: f'(x_0).$$

Es decir, f es derivable en $x_0 \iff \exists L \in \mathbb{R} : \frac{f(x_0+h)-f(x_0)}{h} \xrightarrow{h \to 0} L$.

Definición 2 (Derivabilidad). Sea $U \subset \mathbb{R}$ abierto, $f\colon U \longrightarrow \mathbb{R}$ es derivable

$$\iff \forall x_0 \in U : f \text{ es derivable en } x_0.$$

Referenciado en

- `teo-fundamental-calculo`
- `lem-transformada-fourier-derivada`
- `prop-criterio-dirichlet`
- `clase-schwartz`

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